

Online Test 1 (10%)

- Due 5 Jun at 23:59
- Points 100
- Questions 12
- Available until 5 Jun at 23:59
- Time limit 90 Minutes

Instructions

Due Date: Week 3 Lab Session

Weighting: 10%

About this test

1. This is an online test worth 10% of the total marks for this unit.
2. It consists of 10 questions on theories and applications. Using the lab environment in the quiz is essential to find answers.
3. You may **only attempt this test once**.
4. **Do not navigate away from the test (close the browser tag) before you have completed it and submitted it.**
5. You have **90 minutes** to complete the test; exceeding this will result in a loss of marks and penalties.
6. This is a **close-book test**.
7. Once you begin the quiz, a summary will appear in the upper right corner showing the number of questions (completed and total) and the time remaining.
8. Please bring your student ID to the test and put it on the desk in front of you for the tutor to verify your identity.

This quiz was locked 5 Jun at 23:59.

Attempt history

	Attempt	Time	Score
LATEST	Attempt 1	73 minutes	100 out of 100

⚠️ Correct answers are hidden.

Score for this quiz: 100 out of 100

Submitted 5 Jun at 9:48

This attempt took 73 minutes.



Question 1

9 / 9 pts

Which of the following HDFS Daemon holds actual data in Hadoop?

- ☐ Resource Manager
- ☐ Name Node
- ☐ Master Node
- ☒ Data Node

Lecture 1 - Slide 45



Question 2

9 / 9 pts

Which of the following description is (are) TRUE?

- ☒ HDFS is designed more for batch processing.
- ☐ HDFS is designed more for interactive use by users.
- ☐ HDFS applications need a read once write many access model for files.
- ☒ HDFS applications need a write once read many access model for files.

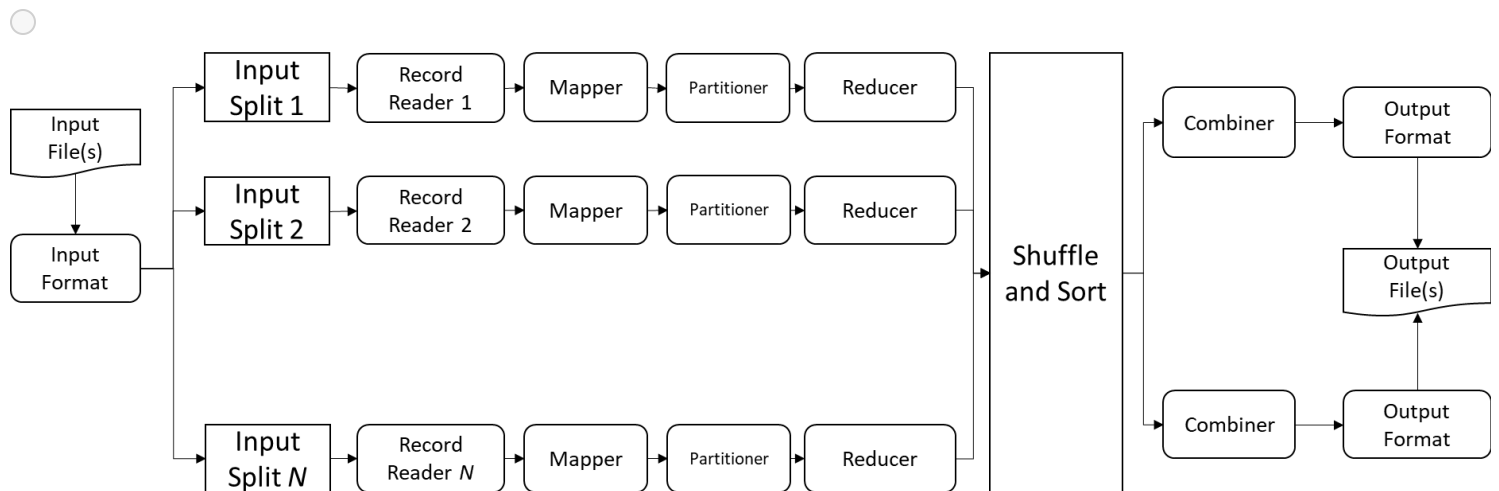
Lecture 1 - Slide 38

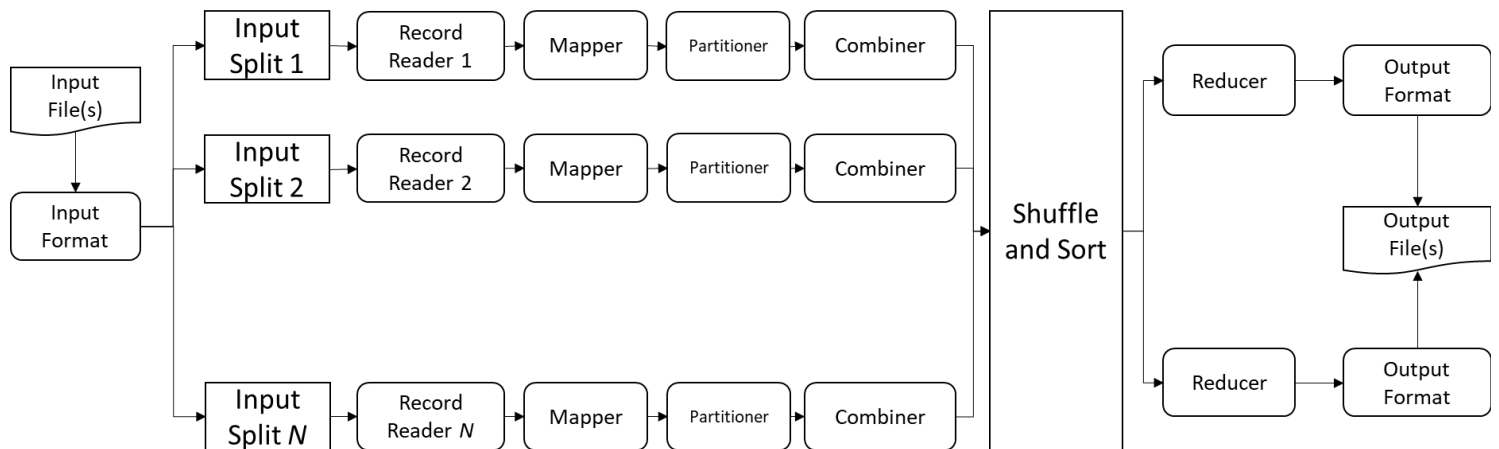
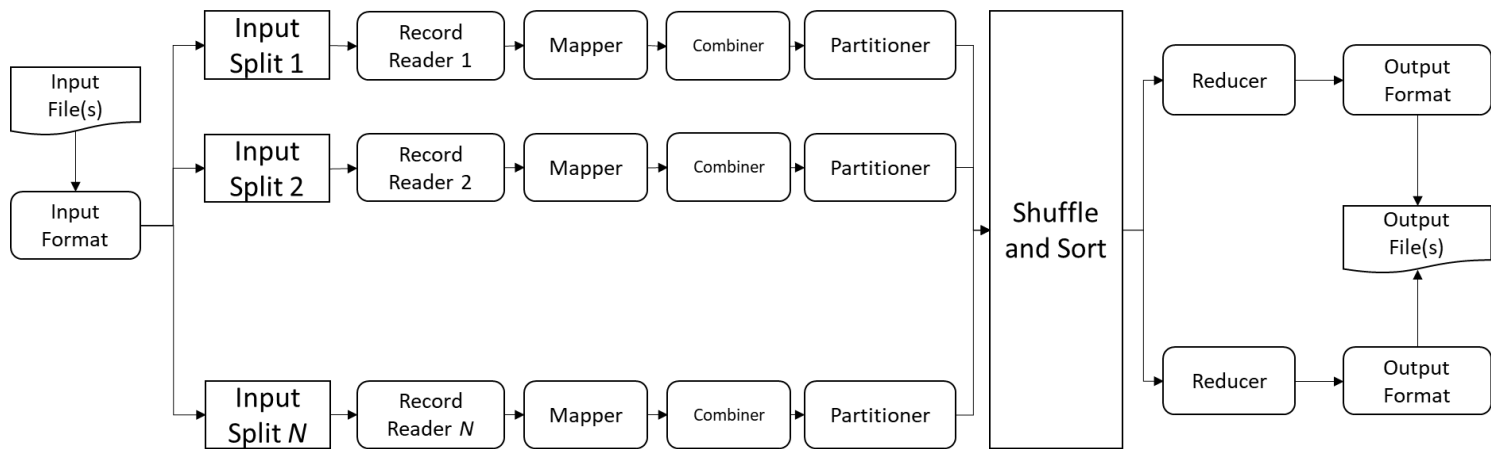
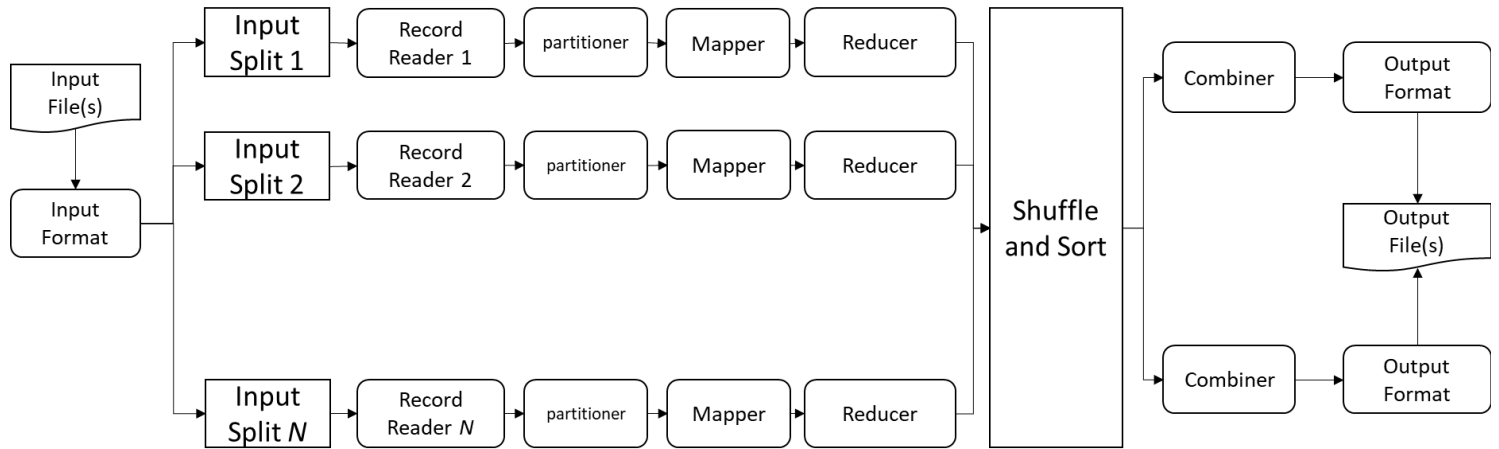


Question 3

9 / 9 pts

Which of the following structure is correct?





Week 03 - Lecture - P5



Question 4

9 / 9 pts

Which of the following description(s) regarding Hadoop is (are) correct?

- ☒ Keys must be WritableComparables because they are passed to the Reducer in sorted order.
- ☐ Keys and values in Hadoop are primitive Java Objects.

- ☒ A WritableComparable is a Writable which is also Comparable.
- ☐ Values are objects which implement Readable.

Week 03 - Lecture - P13.



Question 5

9 / 9 pts

Which of the following description is correct?

- ☒ Apache Pig is a dataflow language.
- ☐ HiveQL can be used to query SQL-based RDBMS.
- ☐ Mahout is a specialised library for supporting massive statistical calculations.
- ☐ SPARK stores intermediate data in multiple data nodes and thus it is much faster than MapReduce.

Lecture 2 - slide 57.



Question 6

9 / 9 pts

What is the full name of HDFS?

- ☐ Hadoop Data-driven File System
- ☒ Hadoop Distributed File System
- ☐ Highly Distributed File System
- ☐ Highly Distributed Functioning System
- ☐ Hadoop Distributed Functioning System

Lecture 1 - Slide 38



Question 7

9 / 9 pts

Which of the following description is correct?

- ☒ The MapReduce job can be submitted by the job object.
- ☐ Hadoop does not have a configuration file. All configuration is defined in the driver code.
- ☐ The job configuration options are not set in the job class but in the mapper class.
- ☐ Hadoop does not have a configuration file.

Lecture 3 - slide 21.



Question 8

9 / 9 pts

Which of the following component in Hadoop eco-system is used to exchange data with RDBMS?

- ☐ Flume
- ☒ Sqoop
- ☐ HBase
- ☐ Oozie

Lecture 2 - Slide 50



Download the source file and open it with any text editor on your computer.

Copy and paste the text content without any modification to the following website:

<https://regex101.com/>  [\(https://regex101.com/\)](https://regex101.com/)

and then answer the question(s) in the next section.

Data Source: [online_test1_application1.txt](#)

<https://swinburne.instructure.com/courses/68676/files/37523336?wrap=1> 

https://swinburne.instructure.com/courses/68676/files/37523336/download?download_frd=1



Question 9

14 / 14 pts

Use the specified tool with regular expressions to find the answers and match them from the left to right.
The count of pattern "them." appears at the end of a line.

18 matches. 

The count of pattern "them." appears.

1524 matches. 

The count of integers in the form with thousands separator (using comma as the separator).

10392 matches. 

The count of integers in the form with thousands separator (comma as the separator) is greater than 999,999.

132 matches. 

Pattern: them\. for finding the count of "them." appears at the end of a line.

Pattern: them\. for finding the count of "them." appears in the article.

Pattern: `\d+\,\d{3}` for the count of integers in the form with thousands separator (using comma as the separator).

Pattern: `(\d{1,3}\,){2,}\d{3}` for the count of integers in the form with thousands separator (comma as the separator) is greater than 999,999.



Finding statistical characteristics of things is an effective way to describe a system.

Based on this idea, we want to find out the average length of each word leading by a particular prefix.

The upper case and the lower case should be distinguished to find more detailed characteristics.

For any text input, your program should report the average length of words that begin with 'a', 'b', and so forth.

For example, if the input is "No now is definitely not the time", the corresponding output would be:

N	2.0
n	3.0
d	10.0
i	2.0
t	3.5

The input file to be used is "shakespeare.tar.gz" located at
"/home/training/training_materials/developer/data" in your virtual machine image.

You are expected to write a MapReduce program and answer the question(s) in the following section.

You can use the Eclipse project "averagewordlength" located at "~/workspace/averagewordlength" in your virtual machine image to complete this task.

Tips:

Here are some simple string operations in Java:

`str.substring(x,y)` gives you the letters starting at position x to y in str

`str.length()` provides you the length of str

If your output looks like the image below, it means that you solve this task correctly.

```
training@localhost:~/workspace/averagewordlength
File Edit View Search Terminal Help
0      6.886792452830188
1      2.152542372881356
2      1.0769230769230769
3      1.0
4      1.5
5      1.3333333333333333
6      1.75
7      1.0
8      1.6666666666666667
9      1.0
A      3.901754225255347
B      5.143532818532819
C      6.634214463840399
D      5.221781152916811
E      5.53018939875429
F      5.265583343912657
G      5.810282153366799
H      4.428398058252427
I      1.4687346778674861
J      4.97845507094062
K      4.659987476518472
L      5.116772823779193
M      5.451585352834419
:
```



Question 10

14 / 14 pts

What is the approximate average length of words that start with the letter "c"?

- ☐ 5.5
- ☐ 8.2
- ☒ 6.1
- ☐ 5.3

Quiz score: 100 out of 100